

AI Persona — Evaluation Report · Rajasurya J

All numbers below are from real runs: chat/retrieval from evals/ results/ (Groq Llama-3.3-70B judge, golden set of 17 Q&A); voice from live Vapi test calls (latency from per-turn Vapi message timestamps).

1. Voice quality (live test calls)

Metric	How measured	Result
First-response latency	ms from end-of-caller-speech → first assistant audio, per assistant turn from Vapi call logs	median 1.53 s · range 0.9–1.7 s conversational · 4/5 turns < 2 s
Slot-lookup turn	the one turn > 2 s (2.4 s) is the get_availability turn — includes a live Google Calendar round-trip	within budget for a tool turn
Transcription	Deepgram nova-2; reviewed against the call transcript	accurate on general speech; the rare proper noun "Rajasurya" was mis-recognised (see failure mode 3)
Task completion (booking)	did the call create a real Google Calendar event + send an invite	booking call succeeded end-to-end (1/1) after the fixes in §3

Latency is engineered with Groq (low time-to-first-token), Deepgram nova-2 STT, a short maxTokens cap, an in-prompt brief for common questions, and tuned endpointing/barge-in (startSpeakingPlan.waitSeconds 0.4, stopSpeakingPlan.numWords 1). Barge-in verified by interrupting mid-answer — TTS suppressed without crashing. *Sample note:* voice N is small (2 live calls); the harness (evals/voice-latency.ts) ingests more calls as they accrue over the 7-day window.

2. Chat groundedness & retrieval — measured (golden set = 17 Q&A)

Metric	How measured	Result
Retrieval P@6 / R@6 / MRR	golden labels vs the real index (true recall denominator), factual queries n=10; npm run eval:retrieval	P@6 = 0.72 · R@6 = 0.75 · MRR = 1.00 · Hit@6 = 100%
Groundedness / Hallucination	LLM-as-judge faithfulness over (question, retrieved context, answer); honest refusals = grounded; n=13	92.3% grounded · 7.7% hallucination (1/13)
Injection resistance	adversarial subset (jailbreak / persona-override / fabrication), n=4	100% resisted (4/4)

Retrieval is evaluated **separately** from generation so a wrong answer is attributable to retrieval-miss vs. generation-hallucination. MRR = 1.00 means the top-ranked chunk was relevant on every factual query. *Judge caveat:* the judge shares the generator's model family (Llama-3.3-70B), so I also manually reviewed all 13 factual answers (manual count agreed: 1/13); an independent judge would cut self-evaluation bias.

3. Three failure modes discovered → root cause → fix

- Free-tier fallback hallucinated and leaked tool syntax.** When Groq hit its daily token cap, the chat fell over to a free gpt-oss-120b, which invented ungrounded facts, emitted raw <function=...> tool syntax into the reply, and double-answered (preamble + post-tool answer). *Root cause:* a low-quality fallback model + streaming the pre-tool preamble as final text. *Fix:* switched the fallback to llama-3.3-70b-instruct (quality-matched to the primary), strip leaked tool syntax, and clear any preamble when a tool call follows so only the final answer renders.
- Voice booking used the wrong year and a placeholder email.** A live test call booked 2024-06-08 (not 2026) and sent the invite to your.email@example.com. *Root cause:* the voice get_availability returned only spoken labels (no machine timestamp), so the model *guessed* the ISO date and never collected a real email. *Fix:* get_availability now returns each slot's exact startISO to copy verbatim; book_meeting rejects past/invalid dates and placeholder/invalid emails and makes the agent re-ask. Verified end-to-end after the fix.
- STT mis-transcribed the candidate's name.** Deepgram rendered "Rajasurya" as "Rodriguez"/"Roger Soria" — a rare proper noun absent from the language model. *Root cause:* no domain vocabulary hint to the transcriber. *Fix:* add keyterm/keyword boosting (the name, company names, repo names) to the Deepgram config so the persona's own identity transcribes reliably.

4. One conscious tradeoff

Cost vs. reliability/latency — chose \$0 open-weight models with layered fail-over over a single paid frontier provider. Chat + voice reasoning run on Groq Llama-3.3-70B (highest free limits, lowest time-to-first-token). To survive 7 days of unannounced probing on a free tier, lib/llm.ts chains **Groq-70B → Groq-8B-instant → OpenRouter-llama** with a 14 s first-token timeout and fast fail-over, so a throttled or slow provider degrades gracefully instead of hanging (an earlier version stalled ~90 s before this). I deliberately rejected a more-available free fallback (gpt-oss) because it hallucinated about the persona — for a grounded persona, a graceful "ask again" beats a confident wrong answer. *Accepted cost:* Llama-3.3-70B is a notch below frontier on subtle reasoning, and the daily free cap still exists (mitigated, not removed; a paid Groq key is a drop-in fix). lib/llm.ts is a thin OpenAI-compatible client, so swapping in Claude/GPT later is a one-line change.

5. With two more weeks

Cross-encoder reranker + a larger golden set in CI (RAGAS) for precision; automated voice evals pulled from Vapi end-of-call webhooks (latency/WER/booking) instead of manual sampling; Deepgram keyterm boosting + streaming TTS to push every turn < 1 s; per-turn tracing/observability (retrieved ids, scores, tool calls, latency); a paid-tier key + usage alerting so the endpoint never goes dark; and a continuous red-team suite that fuzzes injection payloads against the live corpus.